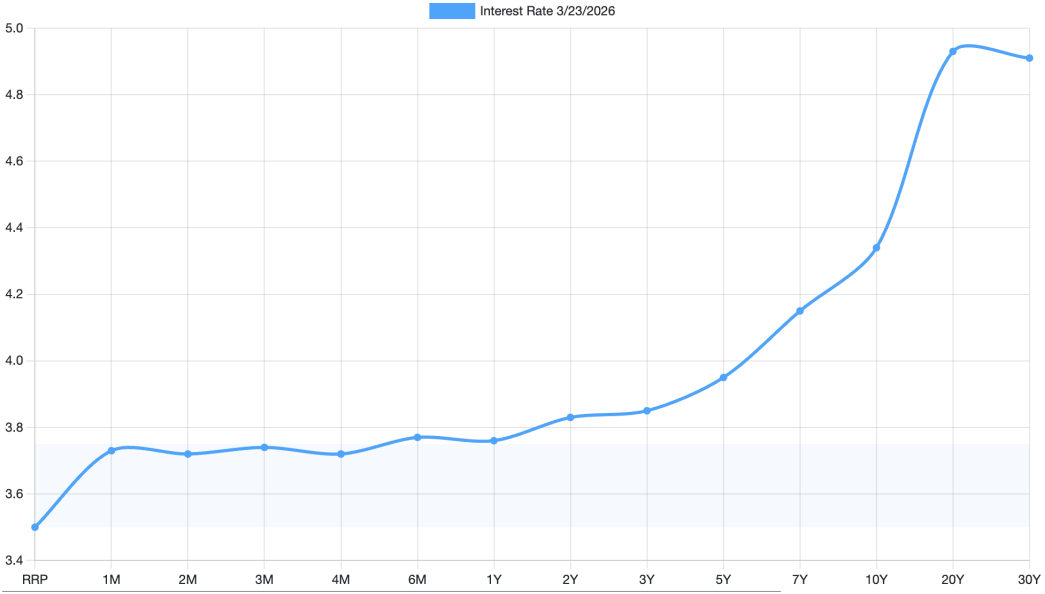
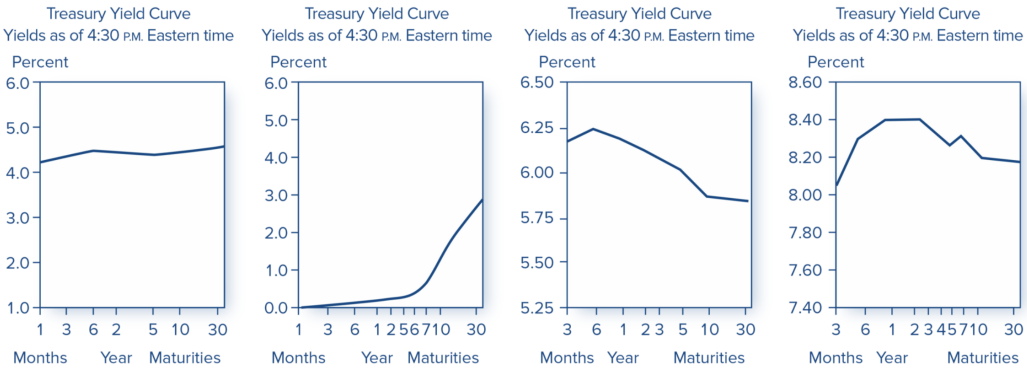
BUS-F420: Equity and Fixed Income Investments

Lecture 16: What is the Yield Curve?

Prof. Derek Wenning

March 25, 2026


Interest Rate 3/23/2026
5.0
4.8
4.6
4.4
4.2
4.0
3.8
3.6
3.4
RRP
1M
2M
3M
4M
6M
1Y
2Y
3Y
5Y
7Y
10Y
20Y
30Y



A. (January 2006) Flat Yield Curve

B. (December 2012) Rising Yield Curve

C. (September 11, 2000) Inverted Yield Curve

D. (October 4, 1989) Hump-Shaped Yield Curve

Where Does the Yield Curve Come From?

- Let's start with a simple world where all future interest rates are <u>known</u>
  - (Will generalize to uncertain interest rates in a bit)
- Suppose the yield curve is upward sloping (e.g. longer maturity = higher yields)
- **What does this imply for future <u>short term</u> interest rates?**

Where Does the Yield Curve Come From?

- Let's start with a simple world where all future interest rates are <u>known</u>
  - (Will generalize to uncertain interest rates in a bit)
- Suppose the yield curve is upward sloping (e.g. longer maturity = higher yields)
- **What does this imply for future <u>short term</u> interest rates?**
  - **Answer:** Rates must be increasing over time!

A No-Arbitrage Condition for Two Strategies

- **Strategy 1:**
  Buy 2-year annual zero $(y_2 = 6\%)$ and hold until maturity
  - **Check:** What is the price?

A No-Arbitrage Condition for Two Strategies

- **Strategy 1:**
  Buy 2-year annual zero ($y_2 = 6\%$) and hold until maturity
  - **Check:** What is the price? $\$1000/(1.06)^2 = \$890$

- **Strategy 2:**
  (i) Buy \$890 worth of a 1-year annual zero with a yield of 5%
  (ii) Roll over proceeds into another 1-year annual zero

A No-Arbitrage Condition for Two Strategies

- **Strategy 1:**
  Buy 2-year annual zero ($y_2 = 6\%$) and hold until maturity
  - **Check:** What is the price? $\$1000/(1.06)^2 = \$890$

- **Strategy 2:**
  (i) Buy \$890 worth of a 1-year annual zero with a yield of 5%
  (ii) Roll over proceeds into another 1-year annual zero

- **In the absence of arbitrage, what does this imply for the 1-year rate at time 2?**

$$\textbf{Stratgy 1 Return} = \textbf{Strategy 2 Return} \qquad \text{(No arbitrage)}$$

$$\$890 \times 1.06 \times 1.06 = \$890 \times 1.05 \times (1 + r_2)$$

$$\rightarrow \mathbf{1 + r_2 = (1.06)^2/1.05 \approx 1.07}$$

What About a 3-Year Strategy?

- **Strategy 1:** Buy 3-year annual zero ($y_3 = 7\%$) and hold until maturity
  - Price?

What About a 3-Year Strategy?

- **Strategy 1:** Buy 3-year annual zero ($y_3 = 7\%$) and hold until maturity
  - Price? $\$1000/(1.07)^3 = \$816.30$
- **Strategy 2:** Buy a 2-year zero and rollover proceeds into another 1-year zero
  - Based on last slide, 2-year yield $= 6\%$
- **In the absence of arbitrage, what does this imply for the 1-year rate at time 3?**

What About a 3-Year Strategy?

- **Strategy 1:** Buy 3-year annual zero ($y_3 = 7\%$) and hold until maturity
  - Price? $\$1000/(1.07)^3 = \$816.30$

- **Strategy 2:** Buy a 2-year zero and rollover proceeds into another 1-year zero
  - Based on last slide, 2-year yield = 6%

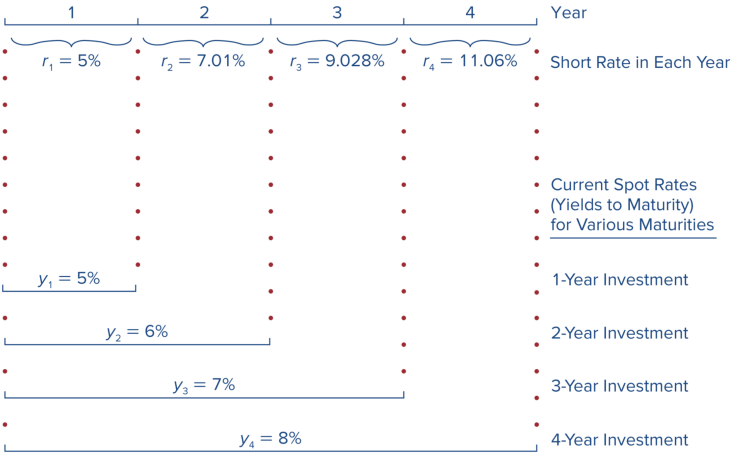
- **In the absence of arbitrage, what does this imply for the 1-year rate at time 3?**

$$\textbf{Strategy 1 Return} = \textbf{Strategy 2 Return} \qquad \text{(No arbitrage)}$$

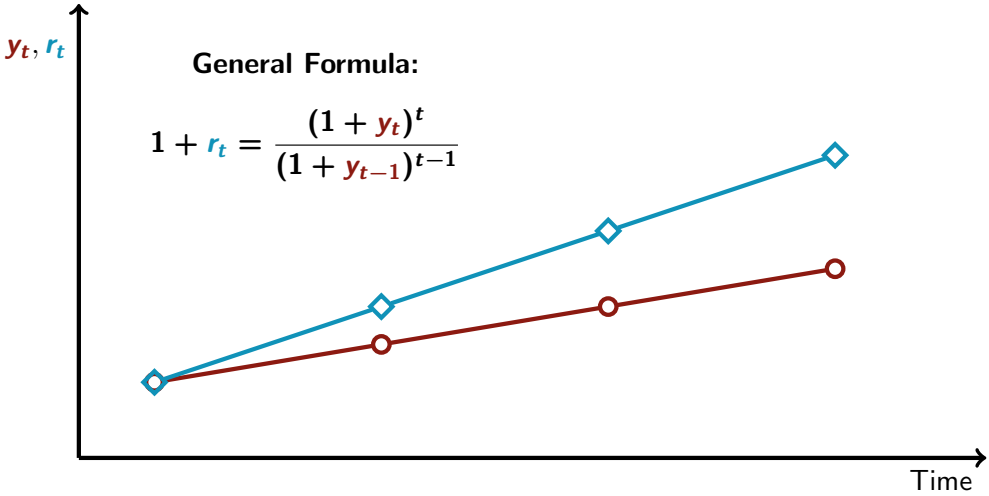
$$\$816.30 \times (1.07)^3 = \$816.30 \times (1.06)^2 \times (1 + r_3)$$

$$\rightarrow \mathbf{1 + r_3 = (1.07)^3/(1.06)^2 \approx 1.09}$$

Visual Representation of the No-Arbitrage Strategy



Yield Curve Implies the Path of All Future Short Rates



General Formula:

$$1 + r_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}$$

Exercise: Calculate Spot Rates Given a Yield Curve

- Let's practice calculating short rates given a yield curve <u>under full information</u>:

$$\textbf{Formula:} \quad 1 + r_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}$$

Maturity	Yield	Short Rate
1	3%	3%
2	4%	
3	6%	
4	7%	

Exercise: Calculate Spot Rates Given a Yield Curve

- Let's practice calculating short rates given a yield curve <u>under full information</u>:

$$\textbf{Formula:} \quad 1 + r_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}$$

Maturity	**Yield**	**Short Rate**
1	3%	3%
2	4%	**5%**
3	6%	
4	7%	

Exercise: Calculate Spot Rates Given a Yield Curve

- Let's practice calculating short rates given a yield curve <u>under full information</u>:

$$\textbf{Formula:} \quad 1 + r_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}$$

Maturity	**Yield**	**Short Rate**
1	3%	3%
2	4%	**5%**
3	6%	**10%**
4	7%	

Exercise: Calculate Spot Rates Given a Yield Curve

- Let's practice calculating short rates given a yield curve <u>under full information</u>:

$$\textbf{Formula:} \quad 1 + r_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}$$

Maturity	**Yield**	**Short Rate**
1	3%	3%
2	4%	**5%**
3	6%	**10%**
4	7%	**10%**

Are These The True Short Rates?

- What we've calculated is a no-arbitrage-inspired short rate
  - **Key Assumption:** all future rates are known with certainty!
  - If this were true, no arbitrage would be a correct concept
- In reality, interest rates are <u>uncertain</u> – can't know for sure if our rates = true rates
- Our short rate is known as the **forward rate** (f_t)
  - e.g. what would prevail without uncertainty

Incorporating Interest Rate Risk and Uncertainty

- Suppose you're a short term investor deciding between two strategies:
  - Buy 1-year zero ($y_1 = 5\%$) then rollover into new 1-year zero
  - Buy 2-year zero ($y_2 = 6\%$) and hold to maturity
  - **Assumption:** You expect true $r_2 = 6\%$

  The Problem: you may be forced to liquidate the position in 1 year
  (Creates exposure to r_2!)

- With no arbitrage, what would y_2 be if beliefs were correct ($f_2 = E[r_2]$)?

$$y_2 = \Big[(1.05) \times (1.06)\Big]^{1/2} - 1 = 5.5\%$$

This is less than observed yields $\rightarrow$ risk premia from potential for liquidation

Incorporating Interest Rate Risk and Uncertainty

- Let's work backwards the other way. Suppose $y_2 = 6\%$ and $E[r_2] = 6\%$.
- As before, this implies that the *forward rate* for time 2 is $\boldsymbol{f_2 = 7\%} > E[r_2]$!
  - *Disconnect* between forward and expected future rates!
- Call this gap a **liquidity premium**:

$$f_2 = E[r_2] + \textbf{liquidity premium}$$

(called a liquidity premium since investors are uncertain about being able to sell long-term bond at a given price in the future)

Exercise: Forward rates with uncertainty

$$1 + f_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}, \qquad f_t = E[r_t] + \text{Liq. Prem.}$$

Maturity	Yield	Expected Short Rate	Forward Rate	Liquidity Premium
1	4%	6%	—	—
2	5%	6%		
3	6%	6%		
4	5%	6%		

Exercise: Forward rates with uncertainty

$$1 + f_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}, \qquad f_t = E[r_t] + \text{Liq. Prem.}$$

Maturity	Yield	Expected Short Rate	Forward Rate	Liquidity Premium
1	4%	6%	—	—
2	5%	6%	**6%**	**0%**
3	6%	6%		
4	5%	6%		

Exercise: Forward rates with uncertainty

$$1 + f_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}, \qquad f_t = E[r_t] + \text{Liq. Prem.}$$

Maturity	Yield	Expected Short Rate	Forward Rate	Liquidity Premium
1	4%	6%	—	—
2	5%	6%	**6%**	**0%**
3	6%	6%	**8%**	**2%**
4	5%	6%		

Exercise: Forward rates with uncertainty

$$1 + f_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}}, \qquad f_t = E[r_t] + \text{Liq. Prem.}$$

Maturity	Yield	Expected Short Rate	Forward Rate	Liquidity Premium
1	4%	6%	—	—
2	5%	6%	**6%**	**0%**
3	6%	6%	**8%**	**2%**
4	5%	6%	**2%**	**-4%**

What Determines the Liquidity Premium?

- The precise value of the liquidity premium comes from **investor time horizons**
  - Already saw that *short-term* investors lead to a *positive* liquidity premium
  - Due to higher risk of liquidiation

- Conversely, what if all investors are long-term investors?
  - Could lock in a 6% rate over 2-years
  - Incur risk from 1-year rates if rolling over 1-year zeros

- **But:** Long-run investors (likely) don't need to liquidate!
  - **What does this imply about the liquidity premium?**

- **Key idea:** liquidity premia are *decreasing* with investors' time horizons

Theories of the Term Structure: Expectations Hypothesis

- One might expect that short- and long-run investors net out in the aggregate
- In this case, **what would you expect liquidity premiums to be?**

Theories of the Term Structure: Expectations Hypothesis

- One might expect that short- and long-run investors net out in the aggregate
- In this case, **what would you expect liquidity premiums to be?**
- If the two investor types net out, should expect $f_2 = E[r_2] \rightarrow$ liquidity premium $= 0$!
  - **Implication:** forward rates are approximations for market participants' expected future interest rates

An Application/Test of the Expectations Hypothesis

- Suppose we calculate forward rates for **nominal** and **real** inflation rates
  - "Real" = TIPS treasuries (inflation-protected)

- **What might we learn from the spread between these forward rates?**

An Application/Test of the Expectations Hypothesis

- Suppose we calculate forward rates for **nominal** and **real** inflation rates
  - "Real" = TIPS treasuries (inflation-protected)

- **What might we learn from the spread between these forward rates?**

$$f_t^{nom} - f_t^{real}$$

An Application/Test of the Expectations Hypothesis

- Suppose we calculate forward rates for **nominal** and **real** inflation rates
  - "Real" = TIPS treasuries (inflation-protected)

- **What might we learn from the spread between these forward rates?**

$$f_t^{nom} - f_t^{real} \overset{(EH)}{=} E[r_t^{nom}] - E[r_t^{real}]$$

An Application/Test of the Expectations Hypothesis

- Suppose we calculate forward rates for **nominal** and **real** inflation rates
  - "Real" = TIPS treasuries (inflation-protected)

- **What might we learn from the spread between these forward rates?**

$$f_t^{nom} - f_t^{real} \overset{(EH)}{=} E[r_t^{nom}] - E[r_t^{real}] \overset{Fisher}{=} \textbf{Expected Inflation}$$

- **Test:** Look at nominal-real forward rate spread around Fed rate cut announcement

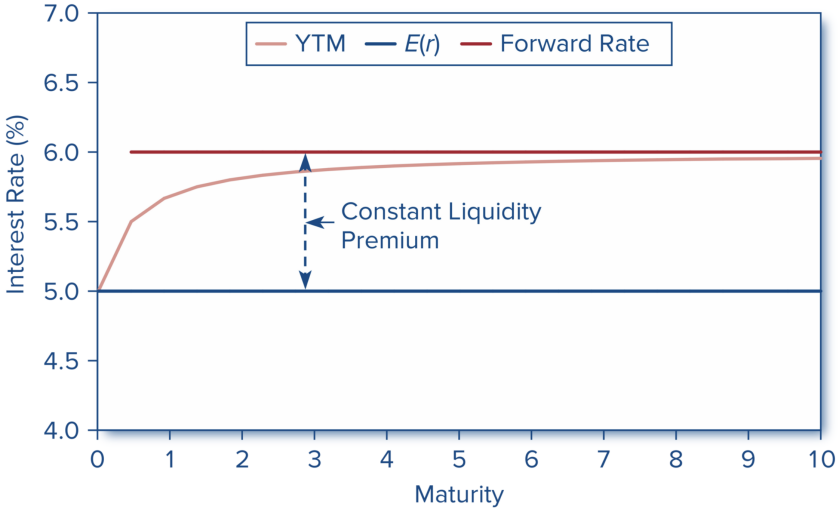
An Application/Test of the Expectations Hypothesis

Theories of the Term Structure: Liquidity Preference

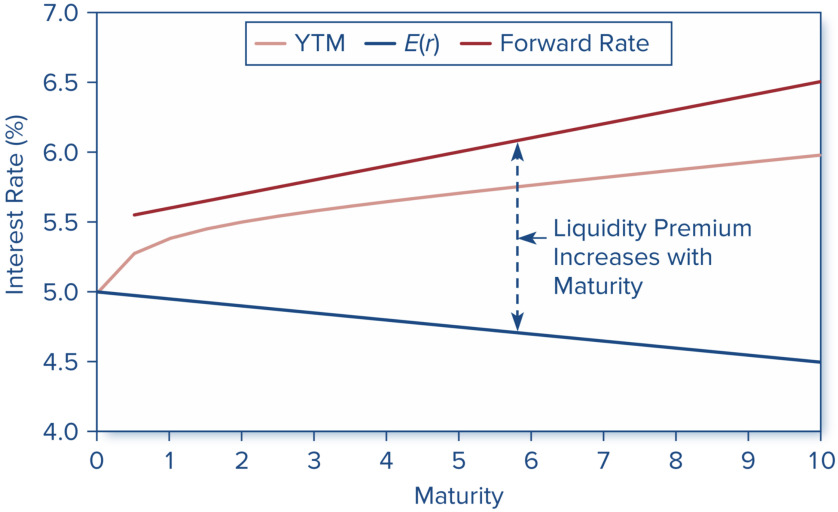
- Of course, the possibility that $f_t = E[r_t]$ is a stringent one
- The **liquidity preference** hypothesis instead states that investors are short-term
  - **Implication:** $f_t > E[r_t]$
- Allows us to *infer* short rates **given an assumption about liquidity premia**
- Given yields y_t and liquidity premia ℓ_t, expected short rates must be

$$E[r_t] = f_t - \ell_t = \frac{(1 + y_t)^t}{(1 + y_{t-1})^{t-1}} - \ell_t$$

Examples of The Liquidity Preference Theory



Examples of The Liquidity Preference Theory



Theories of the Term Structure: Market Segmentation

- Previous theories: investors could **perfectly substitute** across maturities
  - Not always the case!
- May be that the term structure is simply due to **segmented markets**
  - *Example 1:* Money market mutual funds can only invest in very short maturity treasuries
  - *Example 2:* Life insurers have to match maturity of liabilities $\rightarrow$ long-term bonds
- If more short- than long-term investors, simple reason why $y_1 < y_T$: supply and demand
  - But harder to justify when yield curve is inverted!

The Slope of the Term Structure as an Economic Indicator

- Let's return to the expected rate equation when we have a liquidity premium:

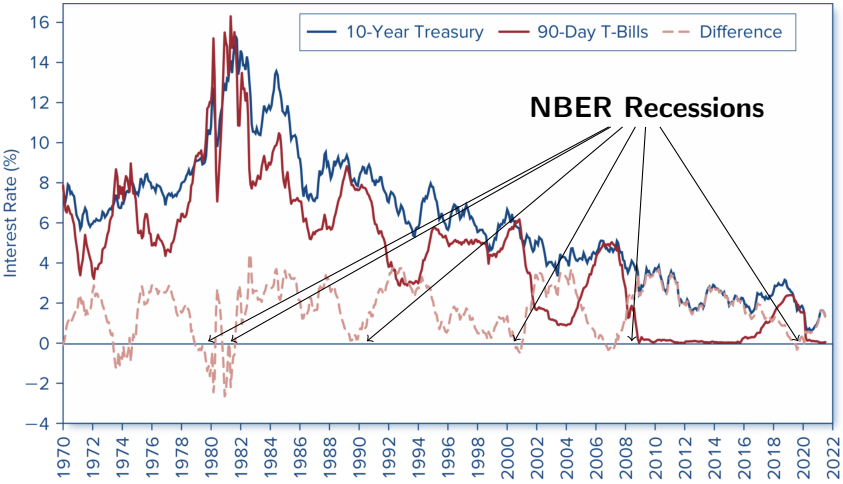
$$E[r_t] = f_t - \ell_t = \frac{(1+y_t)^t}{(1+y_{t-1})^{t-1}} - \ell_t$$

- **Key idea:** Expected future rates move with the *change* (slope) of yields
  - $y_t > y_{t-1}$: $E[r_t]$ is high
  - $y_t < y_{t-1}$: $E[r_t]$ is low

- But why would future rates be low? Fisher hypothesis:
  1. Low future inflation
  2. Low future real rates

$\rightarrow$ **both cases potentially indicative of a recession**

The Slope of the Term Structure as an Economic Indicator



Recap and Next Time

- The term structure of interest rates (e.g. the yield curve) is an important tool for investors
  - Informs about forward rates and expected short rates
  - Practical as an economic indicator
  - Allows us to properly value coupon bonds

- **Next time:** practice problem set